



Deep Reinforcement Learning approach for Small Bodies Shape Reconstruction Enhancement

Margherita Piccinin* and Michèle Lavagna†
Politecnico di Milano, Via La Masa 34, Milano, 20156, Italy

This paper tackles Autonomous Small Bodies Exploration (ASBE), i.e. the problem of mapping an unknown comet or asteroid while autonomously navigating in its proximity. A Deep Reinforcement Learning (DRL) based planning policy is designed to increase mapping efficiency with a smart autonomous selection of the images acquisition times. Such policy is verified with an Image Processing algorithm that generates a point cloud of the body shape model, based on the acquired images, via feature detection and matching.

Nomenclature

ASBE	=	Autonomous Small Bodies Exploration
DQN	=	Deep Q Network
DRL	=	Deep Reinforcement Learning
FOV	=	Field-of-View
IP	=	Image Processing
MDP	=	Markov Decision Process
POMDP	=	Partially Observable Markov Decision Process
SLAM	=	Simultaneous Localization And Mapping
SPC	=	Stereophotoclinometry
SPG	=	Stereophotogrammetry

I. Introduction

AUTONOMOUS Small Bodies Exploration (ASBE) is an arduous problem for future space missions, in relation to the limited on-board resources. In small bodies proximity, the environment is extremely harsh and unknown: operations entail articulated phases, from far approach to close surveys, for gradually characterizing the body with different instruments and techniques. In particular, body imaging is fundamental for the body shape reconstruction, that is performed on-ground with stereophotoclinometry (SPC) [1] and stereophotogrammetry (SPG) [2] techniques. Today, ground support is necessary for navigation and mapping tasks and a large human effort is required even 24/7 for proximity operations supervision and planning [3]. Autonomous operations would reduce the burden of routine navigational support and communication requirements on network services, thus decreasing the mission cost. Autonomy is desirable also to improve the mission science return (high value data), enabling opportunistic science and real-time re-planning, otherwise impossible because of communications delays.

This paper proposes an innovative DRL-based planning for efficient small bodies imaging validating the obtained mapping quality by means of an Image Processing (IP) algorithm that generates a shape point cloud.

In terrestrial robotics, techniques for autonomous exploration have been developed for real applications. In particular, in active SLAM (Simultaneous Localization And Mapping) the robot autonomously localizes itself, maps the environment and plans exploration, tasks that are tightly coupled [4],[5],[6]. Active SLAM has also been examined as being an instance of a Partially Observable Markov Decision Process (POMDP)[7]. In this context, machine learning is a powerful tool that allows to solve otherwise computationally intractable decision processes. Space environment poses additional challenges (chaotic dynamics, low surface reflectivity, variable illumination) and unknowns (body shape, irregular gravity field, etc); so ASBE has never been accomplished. For systematic asteroid exploration, light and robust

*Research Fellow, Department of Aerospace Sciences and Technologies, margherita.piccinin@mail.polimi.it

†Associate Professor, Department of Aerospace Sciences and Technologies, michelle.lavagna@polimi.it.

algorithms are required, capable to promptly react to unexpected conditions, reducing risks for such a delicate phase. Research on this topic proposes navigational filters to estimate relative spacecraft-body state and additional parameters, without considering coupling with mapping and planning [8].

Recently, the general problem of ASBE has been framed as a POMDP, in analogy to terrestrial active SLAM [9]. Up to now only few and simplified solutions have been investigated, quite different from one another, leaving many possibilities unexplored. In [9] the POMDP is reduced to completely observable for designing an orbit selection policy. In [10] the POMDP is tackled using Deep Reinforcement Learning (DRL), overcoming the simplifications in [9], but proposing a direct maneuvering policy, hard for an on-board integration. In [11] a DRL-based planning regards the timing of image collection, allowing safe operations with an easy and flexible approach. In all such works, the planning objective is to better the body imaging. Currently there is still a big open issue: the planning objective model is based on hand-designed functions and the goodness of body mapping has never been verified. Such an issue is typical of DRL approaches: the agent learns according to the provided reward and a reward wrong modeling can lead to undesired behaviors, that may compromise the policy success.

Under the same framework of [11], this paper investigates a DRL-based method to plan images acquisition for body mapping, where the contributions are:

- To develop a DRL-based policy for autonomous decision making of image acquisition times, that increases the efficiency of asteroid mapping and enhances the shape model reconstruction.
- To evaluate the actual benefits for the mapping process. The DRL policy is compared with a benchmark pre-scheduled policy. The method efficiency is evaluated by reconstructing the small body shape with the simulated collected images.

The paper is structured as follows. In Section II the small bodies shape reconstruction problem is presented, highlighting possible improvements of proximity operations to ease this complex process. In Section III a description of the small bodies exploration planning problem in terms of a POMDP is provided, presenting analogies and differences with terrestrial robotic exploration. In Section IV a DRL approach is proposed for images collection optimization during the body mapping operations. The strategy verification method is based on an IP algorithm that process synthetic images. In Section V the DRL planning problem is detailed, with all its modeling aspects. In Section VI the presented results show a successful policy obtained for images selection and finally in Section VII conclusions are drawn.

II. Small bodies shape reconstruction

The mapping procedure of an unknown body is a complex task, that requires several iterations: the shape model is refined during the subsequent observations of the body, until a high resolution model is obtained. Two main methods are used for this purpose: SPC and SPG. In this section insights on shape reconstruction techniques are provided, highlighting possible improvements in mission operations.

A. The shape reconstruction problem

High resolution shape model reconstruction is a complex procedure that can only be performed on-ground, requiring to elaborate considerable amounts of data with several loops.

In particular, SPC is a 3D reconstruction method that descends from shape-from-shading, combining photometry and stereoscopy. The SPC technique was first introduced by Gaskell [1] and is based on the derivation of *maplets* (or *L-maps*): small scale, 3D high resolution maps, centered on landmarks. The creation of L-maps is achieved by estimating landmark positions and camera poses in an iterative process, exploiting a *photometric model*. Surface normals can be estimated considering how light is reflected. Once the maplets are created, they can be assembled together in the global shape model.

Given the landmark position $\mathbf{V}$ and the spacecraft position $\mathbf{W}$, both expressed in in body-fixed frame $\mathbf{b}_i$, let's define:

- the camera frame $\mathbf{c}_i$, centered on the camera and with axes $\mathbf{c}_1$ and $\mathbf{c}_2$ belonging to the image plane and $\mathbf{c}_3$ in the camera principal axis direction.
- the landmark frame $\mathbf{u}_i$ with axes aligned to East, North and landmark position direction, centered in the landmark.

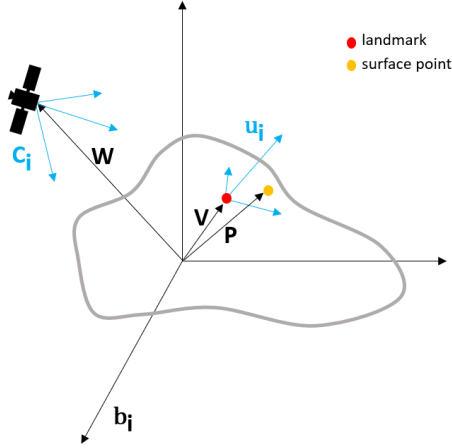


Fig. 1 L-map and camera frames

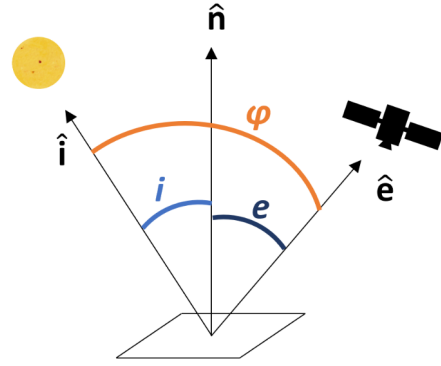


Fig. 2 Angle definition in photometry

A point on the surface with body-fixed coordinates $\mathbf{P}$ can be represented in the L-map frame with coordinates:

$$\begin{aligned} x_1 &= (\mathbf{P} - \mathbf{V}) \cdot \mathbf{u}_1 \\ x_2 &= (\mathbf{P} - \mathbf{V}) \cdot \mathbf{u}_2 \\ h(x_1, x_2) &= (\mathbf{P} - \mathbf{V}) \cdot \mathbf{u}_3 \end{aligned}$$

and in the camera frame coordinates:

$$\begin{aligned} X &= (\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_1 \\ Y &= (\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_2 \\ Z &= (\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_3 \end{aligned}$$

According to the classical pinhole camera model (for instance see [12]), the point can be expressed in homogeneous coordinates and then in the camera coordinate frame in the following way:

$$\begin{bmatrix} X & Y & Z \end{bmatrix}^T \rightarrow \begin{bmatrix} f \frac{X}{Z} & f \frac{Y}{Z} \end{bmatrix}^T$$

where f is the focal length of the instrument. Therefore, the point coordinates in the image are:

$$\begin{aligned} X_1 &= f \frac{(\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_1}{(\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_3} \\ X_2 &= f \frac{(\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_2}{(\mathbf{P} - \mathbf{W}) \cdot \mathbf{c}_3} \end{aligned}$$

The above equations can be rewritten as:

$$X_i = f \frac{(\mathbf{V} - \mathbf{W}) \cdot \mathbf{c}_i + M_{i1}x_1 + M_{i2}x_2 + M_{i3}h}{(\mathbf{V} - \mathbf{W}) \cdot \mathbf{c}_3 + M_{31}x_1 + M_{32}x_2 + M_{33}h} \quad i = 1, 2 \quad (1)$$

where $M_{ij} = \mathbf{c}_i \cdot \mathbf{u}_j$, with $i = 1, 2$ and $j = 1, 2, 3$. So equation 1 shows the geometric relation between the coordinates of a surface point in the maplet frame with its corresponding coordinates in an image taken by the camera.

The photometric model allows to link the brightness of a pixel to the surface slopes and albedo, through a reflectance function R . Given an image k , the brightness I_k of the cell $\mathbf{x} = (x_1, x_2)$ can be predicted with the photometric model:

$$I_k(\mathbf{x}) = \Lambda_k a(\mathbf{x}) R(\phi, e, i) + \Phi_k \quad (2)$$

where a is the relative albedo, normalized to have a unitary mean over the map, Φ_k is the background level and Λ_k is a scale factor that includes conversion from the Sun flux intensity to the pixel signal rate. Please note that the photometric model needs some parameters to be tuned in advance (Φ_k, Λ_k).

The reflectance R of a surface depends on the photometric angles shown in 2 :

- The local angle of incidence i , which is the angle between Sun incidence unit vector $\mathbf{i}$ and surface normal $\mathbf{n}$.
- The local angle of emission e , which is the angle between emission, $\mathbf{e}$ (i.e the spacecraft unit vector) and surface normal.
- The phase angle ϕ , which is the angle between the Sun incidence vector and the emission vector.

Other two angles useful to define the illumination and viewing conditions are:

- The solar azimuth angle α , that is the angle from local North to the incidence vector, projected on the surface plane.
- The spacecraft azimuth angle β , between the emission vector projection on the surface plane and the local North.

Several models of the reflectance function are present in the literature [1, 13]. The reflectance function can be modeled as the weighted sum of Lambert reflectance R_L , that models a pure diffusion reflection, and Lommel-Seeliger reflectance R_{LS} , that models a specular reflection:

$$R = P(\phi) \left[(1 - L(\phi)) R_L(i) + L(\phi) R_{LS}(i, e) \right] \quad (3)$$

$L(\phi)$ is a transition function, $P(\phi)$ represents an exponential decrease with the phase angle. The reflectance models are:

$$R_L(i) = \mathbf{n} \cdot \mathbf{i} \quad (4)$$

$$R_{LS}(e, i) = \frac{\mathbf{n} \cdot \mathbf{i}}{\mathbf{n} \cdot (\mathbf{i} + \mathbf{e})} \quad (5)$$

Defining t_1 and t_2 as the surface slopes respectively along $\mathbf{u}_1$ and $\mathbf{u}_2$, the local surface normal can be written as:

$$\mathbf{n} = \frac{[-t_1, -t_2, 1]^T}{\sqrt{t_1^2 + t_2^2 + 1}} \quad (6)$$

In conclusion, the power of the photometric model consists in the fact that the pixel intensity in an image depends only on the surface slopes and albedo, when the vectors $\mathbf{e}$ and $\mathbf{i}$ are known.

The potential unknowns of the problem are the spacecraft position vector $\mathbf{W}$, the camera orientation $\mathbf{c}_i$, the landmark position and the heights of the map. Two different sub-problems can be identified:

- 1) If an ensemble of landmarks is exactly known for one image, than the camera orientation and the spacecraft location can be retrieved.
- 2) Conversely, if the camera pose is known for several images, the landmark location can be derived.

The global problem can be solved with an iterative method alternatively solving the two sub-problems and with optical data only, except from the scale.

B. Possible improvements of current mapping techniques

The shape reconstruction process can be improved in the autonomy direction. Fundamental inputs to the process are the knowledge of Sun direction and spacecraft pose, related to navigation data, as well as images of the whole body with characteristics that could ease the selected mapping technique. In particular, two frontiers are foreseen to ease on-ground analyses and decrease human support:

- Autonomous navigation with accurate relative pose estimation.
- Image collection optimization, with appropriate illumination and stereo angles.

Both features are also linked to trajectory planning. Of course, depending on the adopted mapping technique different requirements arise; in particular, SPC benefits from large illumination variation. The direction in which the surface is illuminated and observed of course depends on the body shape, that is not completely known. Considering that the spacecraft is exploring an unknown environment, automation and improvement of all these aspects is not a trivial problem.

III. Small Bodies Exploration Planning

THE present section deals with the problem of planning under uncertainty, providing the general framework under which small bodies autonomous mapping falls. In the robotics field autonomous exploration of an unknown environment

is typically formulated with an active SLAM approach, coupling the tasks of mapping, localization and planning. Active SLAM can be seen as an instance of POMDPs. The adaptation of this particular perspective on small bodies exploration is very recent and can be found in literature only in Pesce et al. [9] and Chan and Agha-mohammadi [10] works.

The mathematical formulation of POMDPs is briefly introduced, then the active SLAM problem is presented as a general model for robotic exploration. Finally this model is specifically applied to small bodies autonomous navigation and mapping, highlighting the additional challenges that arise.

A. Partially Observable Markov Decision Processes

Markov Decision Processes (MDP) were introduced by Bellman in the 50s. They are based on Markov chains. A Markov chain is a stochastic process with no memory. This means that given some *states*, the process randomly evolves from one state s_k to another s_{k+1} with a transition probability that depends only on the pair (s_k, s_{k+1}) and not on past states. In a MDP, a decision-maker called *agent* can choose between several possible actions. The transition probability to the next state depends on the chosen action and can be associated to a scalar *reward*. The agent goal is to maximize the rewards over time, with an optimal *policy*. So a MDP is characterized by:

- A state space $\mathcal{S}$.
- An action space $\mathcal{A}$.
- An immediate reward or cost function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{R}$.
- A transition probability $\mathcal{T}(s_{k+1} | a_k, s_k)$, that governs the process by mapping a state-action pair to a probability distribution of states at the next time instant.

Therefore a MDP is a four-tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T} \rangle$. A solution to a MDP is a policy which maps from states to a probability distribution over actions $\pi : \mathcal{S} \rightarrow p(\mathcal{A} = a | \mathcal{S})$. An optimal policy π^* maximizes the reward over time.

The absence of memory in MDPs is defined by the *Markov property*: the next state depends only on the current state and action and not on past actions and states, so the future is conditionally independent of the past, given the present state. This property is essential to many solution algorithms [14]. In real applications the Markov property requirement can be difficult to meet. In order to respect it, the state information must be rich enough so that the observed state transition does not depend on additional historical information. But the agent sensors may not be able to make distinctions between world states. This phenomenon is known as *perceptual aliasing* and can be *involuntary* or *voluntary*. It is involuntary when the sensors can provide only limited or inaccurate information about the environment state. It is voluntary when the agent exploits only part of the sensor information, because of time or resources constraints.

When the state is only partially observable, the problem can be defined as a POMDP. In this case the agent can have only a partial knowledge of the environment: the state is not observable but a signal stochastically related to it is observable. So a POMDP can be described as a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \Omega, \mathcal{O} \rangle$, where:

- $\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}$ are the same spaces defined for the underlying MDP.
- Ω is the space of possible observations.
- $\mathcal{O}(o_{k+1} | a_k, s_{k+1})$ is the probability of making observation o_{k+1} at the next time step, given action a_k that leads to state s_{k+1} .

POMDPs are difficult to be solved, therefore it is better to reduce them in order to find a computationally tractable solution. A POMDP can be reduced to a MDP including the agent history h as internal state. The history is composed by all past actions and observations, so history at time step k will be $h_k = \langle a_0, o_1, a_1, \dots, a_{k-1}, o_k \rangle$. Usually the problem is tackled with a less direct approach known as *belief-space* MDP. This formulation is a tuple $\langle \mathcal{B}, \mathcal{A}, \mathcal{R}_B, \tau \rangle$, where:

- $\mathcal{B}$ is the belief space, with belief $b_k = p(s_k | h_k)$ equal to the probability of being in state s after history h .
- $\mathcal{A}$ is the action space as in the original POMDP.
- r_B is the expected immediate reward $\mathcal{B} \times \mathcal{A} \rightarrow \mathcal{R}$
- $\tau(b_{k+1} | a_k, b_k)$ is the belief transition function, i.e. the probability of reaching the new belief b_{k+1} , starting from b_k and performing action a_k .

The optimal policy is the one that maximizes the reward in the long term, assuming to act according to that policy:

$$\pi^* = \operatorname{argmax}_{\pi} \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}(a_k, b_k) \right] \quad (7)$$

This is called also *infinite horizon* problem, since the reward is maximized over the entire agent lifetime, considering a discount factor $\gamma \in [0, 1]$.

Several approaches exist to solve POMDPs, like methods descending from operations research, point-based algorithms and machine learning [7, 15–18]. POMDPs are often computationally intractable and their complexity

makes it hard to find a solution to their most general formulation. Exact solutions exist only for a very small class of problems formulated as POMDPs.

B. Active Simultaneous Localization And Mapping

SLAM consists in estimating a map of an unknown environment and at the same time localizing in it. Localization is the task of estimating the robot position and orientation (pose) while moving in the environment. This can be done exploiting sensors data of various types (e.g. cameras, laser, sonar), commonly are affected by noise. Moreover as the robot moves, estimation errors are accumulated and this can lead to a divergence from the real pose. The mapping goal is to build a global and consistent map of the environment.

The SLAM problem can be formulated as follows. The environment map at time step k is made of a set of n landmarks $\mathbf{m}_k = \{\mathbf{m}^1, \mathbf{m}^2, \dots, \mathbf{m}^n\}$, where $\mathbf{m}^i$ is the position vector of the i -th landmark. The robot pose $\mathbf{x}_k$ changes while the robot moves under the control $\mathbf{u}_k$ and can be estimated through observations of landmarks location $\mathbf{z}_k$. The agent actions have stochastic effects and the environment is perceived through noisy and partial observations, therefore SLAM can be described in probabilistic terms. If the robot state was perfectly known, the mapping problem would have been governed by probability:

$$p(\mathbf{m}_k | \mathbf{x}_{0:k}, \mathbf{z}_{0:k}, \mathbf{u}_{0:k}) \quad (8)$$

On the contrary, if the map state was known the localization problem would have been:

$$p(\mathbf{x}_k | \mathbf{m}_{0:k}, \mathbf{z}_{0:k}, \mathbf{u}_{0:k}) \quad (9)$$

In practice, the two problems are coupled:

$$p(\mathbf{m}_k, \mathbf{x}_k | \mathbf{x}_0, \mathbf{z}_{0:k}, \mathbf{u}_{0:k}) \quad (10)$$

There are several approaches to solve the SLAM problem, that can be based on filters [19] or on graph optimization [20].

Active SLAM adds to the SLAM problem the planning task [4, 5, 7]. POMDP provides a framework to investigate the effects of actions and observations on the agent's environment perception, thus allowing to design policies that optimize the agent's interaction with the environment in some of its aspects.

Since the environment is stochastic, the problem can be described in probabilistic terms according to the belief-space MDP formulation presented in the previous section III.A. The state vector is composed by robot pose and landmark locations $\mathbf{s}_k = (\mathbf{x}_k, \mathbf{m}_k)$ and its belief is $\mathbf{b}_k = p(\mathbf{s}_k | \mathbf{z}_{0:k})$. The actions that the agent can take coincide with the control $\mathbf{a}_k = \mathbf{u}_k$. The belief is estimated from past control, belief and current observations: $\mathbf{b}_{k+1} = \tau(\mathbf{u}_k, \mathbf{b}_k, \mathbf{z}_{k+1})$. For what concerns the reward, it is usually modeled in terms of an objective function. For instance, a planner can have multiple objectives like maximizing coverage or map accuracy and minimizing navigation time, motion cost or resources utilization. Several criteria exist to formulate the objective function to be optimized by the planning policy [5, 21]. In particular, the exploration problem consists in choosing the sensing trajectory to obtain the best map.

C. Application to space environment

In the present section small bodies navigation and mapping is framed as a POMDP, in analogy with the active SLAM problem. Many similarities can in fact be found between classic robotic and space exploration. In both cases an autonomous robot is exploring an environment that is not completely known and moves in it collecting sensors information. Sensors data are used to localize in the environment and to build a global map of it. On-board resources are limited, both in terms of computational capabilities and energetic resources. Nevertheless, space exploration presents some differences and additional challenges with respect to the classic active SLAM, so the general framework is now tailored on small bodies navigation and mapping.

The relative spacecraft-body state can be defined as $\mathbf{x}_k = (\mathbf{r}_B, \mathbf{r}_B)$ where $\mathbf{r}_B$ and $\mathbf{r}_B$ are the relative position and velocity. Contrary to classic robotics, natural dynamics plays a fundamental role for the state evolution. In addition, the possibility of controlling the motion is constrained by stringent safety requirements and propellant consumption. The sensors used for navigation around small bodies can be of various types, like radar, lidar and optical. The state belief $\mathbf{b}_k$ can be retrieved with determination algorithms based on filters that may exploit also other types of data, coming from proprioceptive sensors (gyroscopes, accelerometers) or star trackers, Sun sensors. In case of optical measurements, the map of the body $\mathbf{m}_k$ is the set of landmarks on the body surface. Landmarks can be generated in different ways, as manually by human experts, with computer vision algorithms for feature extraction and matching or as centers of SPC maplets. The landmarks knowledge is refined during the mission operations and requires human supervision.

Actions represent the ways in which the agent can interact with the environment. In a broad sense, the environment is everything external to the decision maker. Therefore in this case actions can be a change of trajectory or attitude, but also acquisition of sensory inputs or handling of data storage and communication. In principle, there are several objectives towards which the planning goal can be oriented, as optimization of scientific objectives with smart data collection or active SLAM-like autonomous navigation with map accuracy improvement. In practice, it is important to understand which decisions can be taken autonomously without compromising the mission safety. In addition, it has to be considered that spacecraft are complex systems, with which the planning architecture should easily interface. For instance, when planning is used for trajectory design, it should be seen as a high level decision making, letting the task of maneuvering to a suited Guidance Navigation and Control system. The dynamical environment in proximity of small bodies is chaotic and errors in maneuvering can lead to impact or escape from the body. In order to solve the planning problem, it is necessary to reduce the policy space by defining proper states and actions, introducing substantial simplifications to the general POMDP formulation, otherwise computationally intractable. Up to now only few possibilities have been studied. In [9] the agent planning is reduced to a fully observable environment and it is oriented to the maximization of mapping accuracy for SPC, choosing at fixed time intervals an orbit within a subset of possibilities and thus generating an optimal orbit sequence. In [10] the agent directly controls the spacecraft for moving around the body and again benefits imaging for SPC. The agent can also decide in which time steps to take a picture and when to downlink data. In such work state and action spaces are wider and some uncertainty is considered for the spacecraft position knowledge.

IV. Images collection with Deep Reinforcement Learning

This paper tackles the problem of optimizing images collection for small bodies shape reconstruction. The spacecraft acts according to a policy for next best step selection, i.e. for selecting the most proper time instant for collecting a new image, based on the relative pose between camera and body and on the illumination conditions. The decision making problem is solved by means of DRL. A scheme of the proposed architecture for small bodies imaging and shape reconstruction and its verification is schematized in Fig. 3.

The proposed approach comprises two main parts: the autonomous DRL-based decision making and the shape reconstruction through point cloud generation used to validate the DRL policy. Both methods are explained in the following sections.

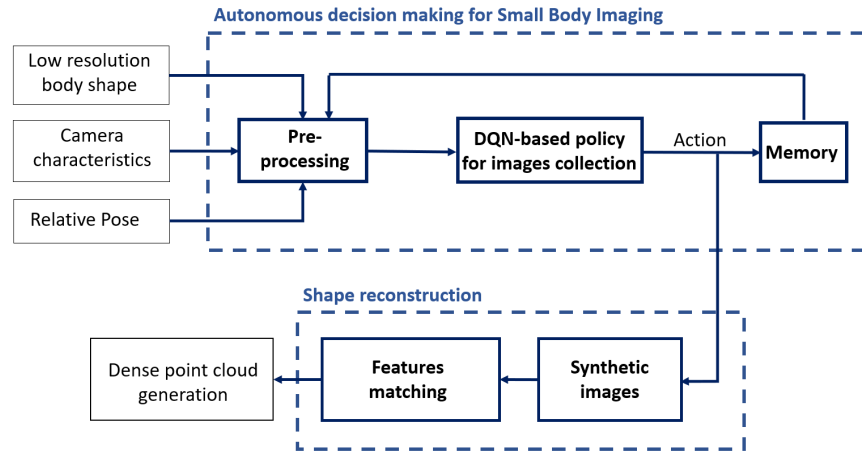


Fig. 3 Autonomous DRL-based planning and shape reconstruction validation method.

A. Deep Reinforcement Learning

Small bodies exploration can be described as a continuous states and actions Partially Observable Markov Decision Process POMDP:

$$\pi^* = \underset{\pi}{\operatorname{argmax}} \mathbb{E}_{\pi} \left[\sum_{k=0}^T \gamma^k r_k(\mathbf{a}_k, \mathbf{b}_k) \right] \quad (11)$$

where γ is the discount parameter, $\mathbf{b}_k$ is the belief of spacecraft and map states $\mathbf{s}_k = (\mathbf{x}_k, \mathbf{m}_k)$, $\mathbf{a}_k$ the actions performed by the agent following policy π and r_k the reward that models the planner objectives.

POMDPs are usually computationally intractable, therefore a reduction of state, action and policy spaces is needed. In the present work, the interaction between agent and unknown environment is represented by the decision of taking a picture of the body at a certain time instant. So the action at the k -th time instant is boolean. The state space comprises a mapping-campaign state, photometric angles of the current frame and on-board memory (based on downlink and imaged still stored on-board). These informations are derived from the knowledge of spacecraft pose, camera field of view, illumination conditions and recorded actions. Therefore they are all available to the agent, that may have a more accurate knowledge of its pose relative to the body than the one available on ground at that moment. A rough body polyhedral shape model is assumed to be available. This assumption is realistic, considering that the body shape is refined in the course of typical mission operations.

This planning framework is different from other research works under many aspects. The here proposed architecture differs from [9] because of different planning goals, less restrictive assumptions and solution method. In fact [9] mainly deals with orbit selection and reduces the environment to completely observable: the planner evaluates reward for a subset of orbits and picks the best. Observations are considered equally spaced on the orbits. It is also interesting to see differences and similarities between the present thesis and Chan's work [10]: a similar solution approach (DRL) is adopted, but the planning goal and model definition are different. In [10] the neural network tasks include not only observations timing, but also downlink and maneuvering: these complex objectives make it difficult and risky to integrate such an architecture on-board. In particular, learning maneuvering introduces issues related to the modeling of the dynamical environment, which are not tackled. Here a safer planning is proposed, that has a large flexibility and generality. The results of this thesis are supported by extensive tests, not carried out in the other works, and the presented architecture is applicable to a wide variety of scenarios.

The POMDP is tackled with DQN [22]: the use of a DRL technique allows to find a good solution policy of an otherwise computationally intractable problem. An Artificial Neural Network (ANN) is used to approximate the Q-value, i.e. the expected return over time. Therefore the optimal policy is the one that maximizes the Q-value:

$$\pi^* = \underset{\pi}{\operatorname{argmax}} Q_{\pi}(\mathbf{a}_k, \mathbf{s}_k | \theta_{\pi}) \quad (12)$$

where θ_{π} are the ANN weights and biases. The ANN architecture is kept light for an low on-board computational time (a multilayer perceptron with 2 hidden layers of 10 neurons each is used).

In [11], DQN has been compared to a simpler DRL technique (NFQ) and to benchmark policies in the same small bodies mapping framework. DQN has proved to have a better performance compared to the other techniques, in several mission scenarios. Nevertheless, the policies have been compared with performance indices and the actual benefits to the mapping process have never been verified. Such aspect is of capital importance since the reward has been modeled based on hand-designed functions.

B. Point cloud generation

Because of the need of verifying the mapping efficiency, a simple IP algorithm is implemented. The policy is meant to optimize images collection for SPC, nevertheless SPC is a very complex and iterative process, usually performed with human experts supervision and its implementation is beyond the purpose of this paper.

A feature detection and matching IP algorithm is implemented for generating a point cloud of the body. According to the actions performed by the agent, the corresponding synthetic images are generated. The spacecraft position and orientation is assumed to be known. A reference image is considered along with images that overlap with the reference frame. SIFT features are extracted and correspondences in the images are searched for: when two features match, based on a Sum Squared Differences metric criterion, the point is triangulated [2]. In such a way a point cloud is created and the body shape reconstructed. The obtained shape is then compared with the available high-resolution shape model, used for generating the images and taken as reference ground truth. Please note that the spacecraft pose is assumed to be known, therefore navigation uncertainty is not accounted for. To decouple the problem is here useful as first step to assess the effectiveness of the images collection policy. Such triangulation-based IP wants to be a verification method rather than a shape reconstruction algorithm to be used during mission operations.

This IP algorithm has two main purposes:

- to assess the effectiveness of the DQN technique;

- to compare the performance of DQN and benchmark policies, using a criterion that overcomes the limitation of having hand-based rewards.

V. Planning problem model for proposed architecture

A detailed definition of the here proposed reduced spaces of the decision process is presented in the next sections, in terms of environment, reward, states and actions.

A. Environment model

The environment here considered is *model-based*, since a model to describe it is available. This model includes all the elements relevant for the mapping campaign. In particular, to derive images acquisition conditions and body coverage, the following elements need to be considered:

- Body geometry.
- Sun direction, relative to the body.
- Camera position and orientation, relative to the body.
- Camera FOV.
- History of collected images conditions.

The body geometry is here modeled with a low resolution polyhedron, made up of triangular facets. Each facet is characterized by a normal unit vector $\hat{n}$, with respect to which photometric angles can be computed (please refer to Figure 2).

The spacecraft and Sun trajectories relative to the body surface are modeled as a set of discrete position vectors. To make an accurate and long term dynamic model of the asteroid approach is beyond the scope of this thesis. The considered small bodies are assumed on a keplerian orbit around the Sun and uniformly rotating about their principal inertia axis. For what concerns the spacecraft dynamics, only the central body attraction is considered (two body problem assumption). The adopted simplified model does not want to represent all possible dynamics environments, that are countless. More complex models would for sure be more realistic, but not more representative, since they would fit only for a very restricted class of mission scenarios. It is therefore believed more useful to adopt here a simple model, but to structure the planning architecture to be effective whatever the relative dynamics between spacecraft and small body. For all these reasons a simplified dynamic model is more convenient and sufficiently realistic to make considerations on image collection planning.

The camera is assumed always pointed towards the body center, without attitude changes. The camera FOV is simplified as conical.

B. Reward model

To appropriately model the reward is a delicate task, because the final policy will maximize what the designer has indicated as valuable: only if the reward is well defined the policy will be able to actually achieve the desired results. Several objectives are accounted for in the reward definition. In particular, the main goals to be achieved are:

- High quality and global map of the small body.
- Fastening of the mapping process.
- Reduction of the amount of data to be downlinked.

The first goal depends on the adopted mapping technique. In the present thesis, reward is defined to ease SPC, but if a different shape reconstruction technique is used, different scores can be defined without changing the overall procedure. Please note that some objective are in contrast and need to be balanced: a planner that maximizes the coverage would explore new areas, while a planner that maximizes the map quality would spend more time on the same areas. In addition, some constraints are present on the number of images that can be taken, because of memory and telecommunication issues. Therefore, only pictures containing relevant information should be collected and sent to ground.

In this section the facet score and facet mapping index used to model the map quality are described. Then, the overall reward is defined, including all the above described objectives.

1. Facet score and facet mapping index

Given a set of images that include a surface portion of the body, a score can be defined to assess the mapping quality for that area. The goodness of the mapping depends mainly on three factors: the illumination conditions, the camera poses and the surface topology. SPC benefits from images with large variations in illumination and small variations in

view angle. So the same surface portion, i.e. the same facet of the representative shape model, should be observed several times under the proper conditions, that are now modeled. The map quality is represented by means of a low resolution shape model, in which a score S^i is associated to each facet i at the considered time instant. This score is the result of the combination of five contributions: incidence, emission, emission variation, solar azimuth angle and spacecraft azimuth angle scores. The scores for SPC here defined rely on previous works [9], with some slight changes.

Incidence score The incidence angle i should be kept between $20^\circ - 60^\circ$ to avoid shadows and excessive brightness, that won't allow the extraction of useful information. Let's define the incidence score S_i^i :

$$s_i = \begin{cases} 1 & \text{if } 20^\circ \leq i \leq 60^\circ \\ \frac{1}{10}i - 1 & \text{if } 10^\circ \leq i \leq 20^\circ \\ -\frac{1}{10}i + 7 & \text{if } 60^\circ \leq i \leq 70^\circ \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

$$S_i^i = \mu_j(s_i) \quad (14)$$

where μ_j is the mean performed over all the n taken pictures that contain the facet.

$$\mu_j(x) = \frac{1}{n} \sum_{j=1}^n (x_j) \quad (15)$$

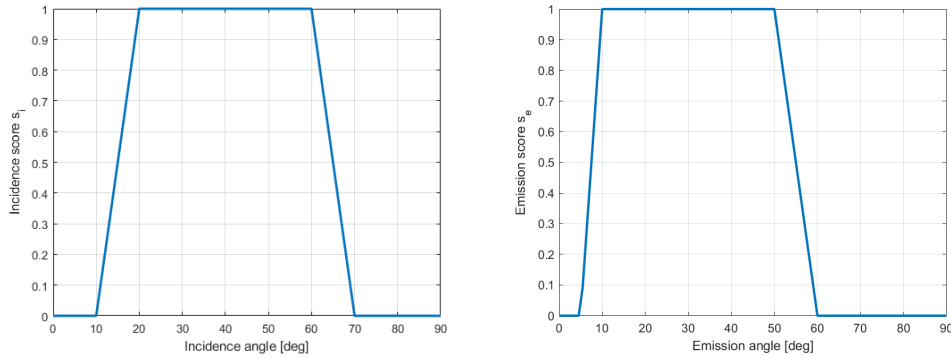


Fig. 4 Incidence and emission scores trend

Emission score The emission angle should be kept between $10^\circ - 50^\circ$. So in a similar manner the emission score S_e^i is defined as follows:

$$s_e = \begin{cases} 1 & \text{if } 10^\circ \leq e \leq 50^\circ \\ \frac{1}{5}e - 1 & \text{if } 5^\circ \leq e \leq 10^\circ \\ -\frac{1}{10}e + 6 & \text{if } 50^\circ \leq e \leq 60^\circ \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

$$S_e^i = \mu_j(s_e) \quad (17)$$

The trend of emission and incidence scores for a single angle value (s_i and s_e) is shown in Figure 4.

Emission variation score Also, a large variation of emission angles is considered beneficial, therefore the emission variation score is:

$$S_{\Delta e}^i = \mu_j \left(\max_k \frac{2\Delta e_{jk}}{\pi} \right) \quad (18)$$

where $\Delta e_{jk} = |e_j - e_k|$. So for each emission angle e_j under which the i -th facet is seen, the maximum difference between the considered angle e_j and the all the other angles e_k under which the facets was observed is computed. Then all the maximum differences are normalized of $\frac{\pi}{2}$, i.e. the maximum possible emission variation, and the mean is performed.

Solar and spacecraft azimuth score Finally, the variation of solar azimuth angles α should be large and the one of spacecraft azimuth angles β small. The respective scores are computed in a similar fashion.

$$S_{\Delta\alpha}^i = \mu_j \left(\max_k \frac{\Delta\alpha_{jk}}{\pi} \right) \quad (19)$$

$$S_{\Delta\beta}^i = 1 - \mu_j \left(\max_k \frac{\Delta\beta_{jk}}{\pi} \right) \quad (20)$$

Please note that in this case the normalizing value is π . All the angle variations are defined as shown in 5.

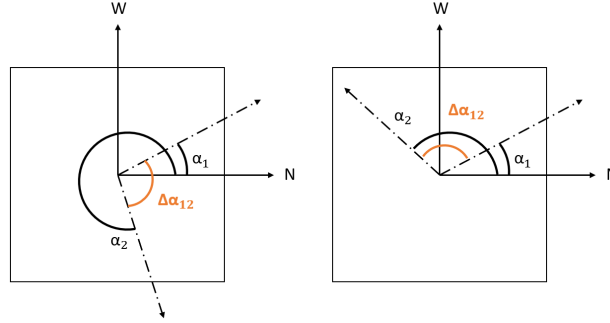


Fig. 5 SPC facet angles variation examples, N = local North, W = local West

Facet score The overall facet score is given by the weighted sum of the different contributions. Please note that the scores defined above have values ranging between 0 and 1.

$$S^i = w_i S_i^i + w_e S_e^i + w_{\Delta e} S_{\Delta e}^i + w_{\Delta\alpha} S_{\Delta\alpha}^i + w_{\Delta\beta} S_{\Delta\beta}^i \quad (21)$$

The weights of the different criteria are computed according to their relative importance following an AHP procedure [23]. The obtained weights are:

$$w_i = 0.56 \quad w_e = 0.17 \quad w_{\Delta e} = 0.06 \quad w_{\Delta\alpha} = 0.10 \quad w_{\Delta\beta} = 0.10 \quad (22)$$

Facet mapping index The overall score represents the goodness of the images, but does not account for the fact that increasing the number of pictures, the knowledge of the area better. Therefore the facet mapping index m^i is defined for the i -th facet:

$$m^i = S^i \min \left(1, \frac{n}{N} \right) \quad (23)$$

where N is an arbitrary number of images. For instance, if $N = 3$, having 3 photos with an ideal maximum score equal to 1 would be satisfying for mapping the area. When the number of images is too low, the area is not well known so an high score is not significant. On the other hand, having at least N images, additional images increase the map knowledge only if they better the score.

2. Reward definition

The immediate reward depends on both states and actions:

$$r_k = r_k(s_k, a_k) \quad (24)$$

If no action is taken the reward is null. When an image is collected in a forbidden state $s \in S^-$, a negative reward equal to -1 is returned to the agent and the image is not accounted for in the successive mapping. Forbidden states correspond to situations in which the image is in complete shadow or when the ideal number of images is overcome, causing problems in on-board data storage. The ultimate goal is to maximize the mapping index, therefore if the picture is taken in allowed states the reward is:

$$\tilde{r} = \mu_m \left(\frac{m_k^i - m_{k-1}^i}{m_k^i} \right) \quad (25)$$

where m_k^i is the mapping index of facet i at time k and μ_m stands for the mean over all the facets in the current frame. Summarizing, the overall reward is:

$$r_k = \begin{cases} -1 & \text{if } a_k = 1 \text{ and } s_k \in S^- \\ 0 & \text{if } a_k = 0 \\ \tilde{r} & \text{otherwise} \end{cases} \quad (26)$$

If the agent immediately takes all the photos that can be sent on ground, for all the successive time steps it will be forced to accept a zero or a negative reward. On the other hand, the long term reward will be higher if images are collected only when it is worth. This formulation of the overall reward is inspired by [24], where prohibited states are defined for modeling learning tasks in control problems.

C. Actions and states model

1. Actions

Here the reduced action space is defined. The agent interacts with the environment only by choosing its sensing locations, so by collecting images, without controlling its relative pose with respect to the body surface. A direct and continuous control $\mathbf{u}_k$ on the trajectory, as the one included in active SLAM problems, would be too risky for this application. As a straightforward consequence of the reduced problem definition, actions are discrete. The action at time step k is simply boolean:

$$a_k = \begin{cases} 0 & \text{if no picture is taken} \\ 1 & \text{otherwise} \end{cases} \quad (27)$$

The number η of pictures to be ideally taken in a certain storage time $T_{storage}$ is fixed. After this storage time images are downlinked and therefore the memory is empty again. The discrete time steps in which an action can be taken are defined in number equal to the ideal number of images times the control parameter Δ_c . Ideally with a large control parameter the final performance would be better, but also the number of decisions to be taken would be too high. Shortening the control interval the overall on-board computational time increases and also the learning time. So a trade off between performance and learning must be done when choosing Δ_c .

2. States

Also the state space is not equal to the complete space of the general POMDP: spacecraft position and orientation, Sun illumination and map representations are reduced. In particular, to directly consider the full and complete map state $\mathbf{m}_k$ would be too onerous, given the large number of landmarks on the body surface. States have been designed to synthesize only the information necessary and useful for decision making. The use of statistical quantities (mean and standard deviation) is the only solution that allows to keep the number of observed states constant despite of the change of number of facets in view. Moreover, to understand how the mapping campaign is proceeding, all the history of past actions should be part of the states as well. Of course to include the whole history in the states observation is not possible, but anyway the POMDP is reduced by making part of the history observable.

The reduced state includes memory, map and angles states, defined as follows:

$$\text{memory state} \begin{cases} s_1 = \left(t - \text{floor}\left(\frac{t}{T_{\text{storage}}}\right)T_{\text{storage}}\right)/T_{\text{storage}} \\ s_2 = \frac{n}{\eta} \end{cases} \quad (28)$$

$$\text{map state} \begin{cases} s_3 = \frac{A_{\text{light}}}{A_{\text{view}}} \\ s_4 = \mu_m(m^i) \\ s_5 = \sigma_m(m^i) \\ s_6 = \mu_M(m^i) \\ s_7 = \sigma_M(m^i) \end{cases} \quad (29)$$

$$\text{angles state} \begin{cases} s_8 = \mu_m(s_i) \\ s_9 = \mu_m(s_e) \\ s_{10} = \mu_m\left(\max_j \frac{\Delta\alpha_{jk}}{\pi}\right) \\ s_{11} = \mu_m\left(\max_j \frac{\Delta\beta_{jk}}{\pi}\right) \\ s_{12} = \mu_m\left(\max_j \frac{2\Delta e_{jk}}{\pi}\right) \end{cases} \quad (30)$$

Memory state The memory state provides information on the time lapse and number of collected images. The idea is that in a certain time interval T_{storage} pictures can be stored in the on-board memory before being sent on ground. The ideal number of images to communicate at every time interval is η . In particular, s_1 represents the percentage of time spent in the current storage interval, while s_2 the number of pictures taken n with respect to the ideal number η . The T_{storage} and η parameters can be tuned depending on mission constraints without affecting the algorithm. These inputs help in evaluating how the collection of a new image would impact on data storage.

Map state The map state provides general information on the mapping campaign advancement. s_3 is the fraction of area in light of the surface portion in view, thus telling the area percentage whose knowledge will actually be improved by a new picture. It can be roughly computed as the ratio between the image facets in light and the total number of facets visible in the image. s_4 and s_5 are the mean of the mapping index and its standard deviation over the surface in view, while s_6 and s_7 are the same quantities over the whole body. These data are useful to decide whether the exploration of the area under exam is worth from the coverage point of view.

Angles state The angles state gives local information about photometric angles under which the facets in view and light are seen at the present time. In particular, s_8 and s_9 are the inclination and emission scores mean. Please note that the means are performed over all the facets in view and in light for the angles of the current time instant only. While s_{10} , s_{11} and s_{12} are the facets mean of the maximum variation of current Sun azimuth, spacecraft azimuth and emission angles with respect to the angles of already take pictures. These inputs allow to evaluate the possible improvement of SPC for what concerns stereo angles and illumination conditions.

Please note that all the states can be estimated on-board through observations of the environment elements described in section V.A. So they can be computed exploiting data that are available to the agent. In particular, during the global mapping phase it can be fairly assumed to have a knowledge of a rough shape model of the body. The camera field of view is of course known, while Sun direction and camera pose can be estimated. In addition, all the states are here assumed to be known with certainty, so the belief is about equal to the states $\mathbf{b}_k \sim \mathbf{s}_k$, therefore simplifying the POMDP. In particular, the memory data are known with certainty, while relative position, attitude and surface illumination can be estimated through sensors and determination algorithms that are typically available on-board.

It has to be highlighted that history of past observations is included as an input since historical information about conditions under which past images were taken is necessary for decision making. This step is useful to eliminate the dependence of the environment state from any historical information not included in the state itself and therefore to respect the Markov property. Nevertheless, to input the full history to the neural network is not possible and data about past photometric angles and mapping quality need to be preprocessed. This implies a loss of information that makes the environment not fully observable, similarly to voluntary perceptual aliasing.

With the presented definitions of reward, states and actions, the general POMDP is reduced to:

$$\pi^* = \operatorname{argmax}_{\pi} \mathbb{E}_{\pi} \left[\sum_{k=0}^T \gamma^k r_k(\mathbf{a}_k, \mathbf{s}_k) \right] = \operatorname{argmax}_{\pi} Q_{\pi}(\mathbf{a}_k, \mathbf{s}_k | \theta_{\pi}) \quad (31)$$

Please note that the choice of DRL as solving tool to design the policy perfectly fits this application. In fact, DRL can well handle large state and observation spaces, but only discrete and low dimensional action spaces.

3. States computation

In order to feed the states to the neural network, a preprocessing block is included in the algorithm (see Figure 3). The preprocessing block has the aim of elaborating known environmental information for the generation of the neural network inputs. The main difficulty is how to exploit them without an exceeding computational cost. At each considered time instant, the facets in the camera field of view and in light are detected. This can be easily done with geometrical considerations only, thanks to the knowledge of the polyhedral shape model, the camera pose and the Sun position relative to the surface. Of course it is necessary to account for the body self-occlusion and self-shadowing. In fact, having small bodies irregular shapes, some surface areas may be not visible because hid by other portions of the body and in a similar manner, the shadow distribution on the surface is influenced by the shape irregularity. Figure 6 shows the relevance of self-occlusion and self-shadowing for an irregular body.

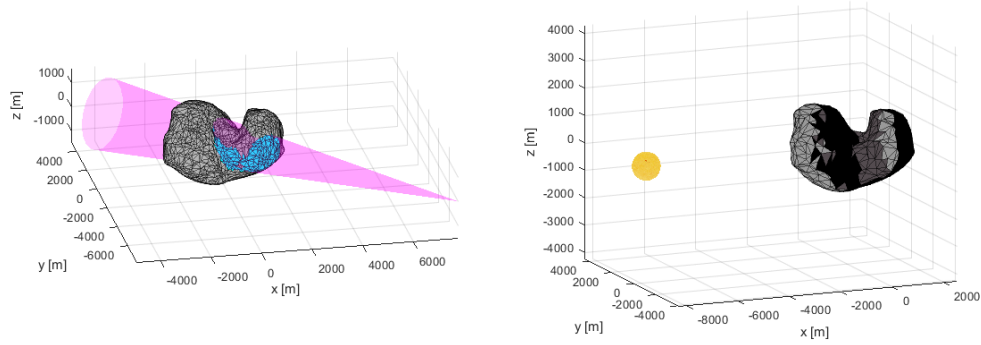


Fig. 6 Relevance of self-occlusion and self-shadowing - 67P-CG shape model.

Then, for the facets in view and in light, the photometric angles are computed. At this point, the computation of the neural network inputs is straightforward. The only historical information needed are the photometric angles under which each facet was seen in past taken images. In particular, photometric angles are stored for each facet and facets scores and mapping index are updated.

Please note that the preprocessing block is the one that has the largest computational cost, mainly because computations related to self-shadowing and self-occlusion involve a double loop on the shape model facets. Anyway this step is fundamental to increase the algorithm generality and to synthesize information provided to the network. It would also be possible not to consider the shape model directly and to implement a different algorithm architecture that does not need to preprocess geometric data, thus saving computational time. Anyway this would open many other issues, like how to handle historical information and how to represent the map state. Moreover, if geometrical information is not included in the network inputs, the task of relating camera pose, illumination and surface geometry would be transferred to the network and needed to be learned during the training. Such approach implies an increased complexity and a loss of generality: the mapping quality is strictly related to the surface geometry and there is no way to deduce it if the training has been done on a different body. So, the present approach prefers to accept the largest computational cost and critically investigate the consequences of a delay in the decisions, while keeping a large generality and independence from the specific mission.

VI. Results

As a target for testing the images collection DRL-based policy, comet 67P has been chose in virtue of its irregular shape. The orbital scenario is a keplerian hyperbolic arc with pericenter at 70 km from 67P, shown in Fig. 7. A

narrow angle camera with a 5° field of view and 1024×1024 pixels is considered as optical sensor. Therefore the body characteristic dimension is about 800 pixels and the resolution with which the surface is imaged is 7 m per pixel. The phase angle Sun-body-spacecraft ranges from 10° to about 90° along the orbit, granting an appropriate illumination for the body imaging and at the same time allowing sufficiently variable illumination conditions for testing the algorithm in a challenging scenario.

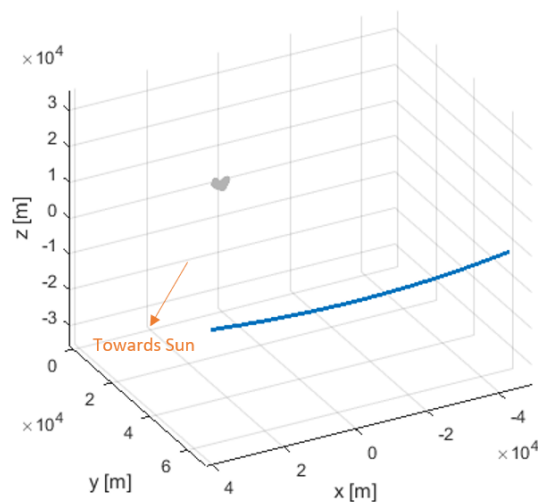


Fig. 7 Orbital scenario - hyperbolic arc at 67P

Since the body is still far, the aimed number of images is assumed to be 30 and the preliminary knowledge of the shape model is assumed to be quite low, as represented in Fig. 8, corresponding to a 100 facets shape model.

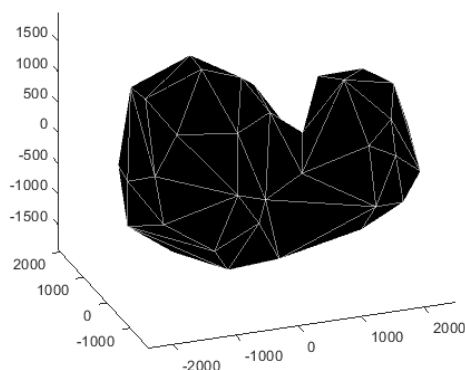


Fig. 8 Low resolution shape model available on-board

The collected images are synthetically generated with the ray-tracing software POV-Ray and the IP algorithm is applied to reconstruct the shape. SIFT features are extracted from images of the same body areas and matched (see Fig. 9).

The DRL-based policy is capable of dealing with memory constraints, collecting a total number of 33 images (slightly larger than the aimed number of 30 images). The point clouds of the UNI and DRL policies are shown respectively in Fig. 10 and 11. Following the DQN policy the mapping quality is better and excluding outliers 938 points are triangulated, whereas with the UNI policy only 162 points are correctly matched and triangulated.

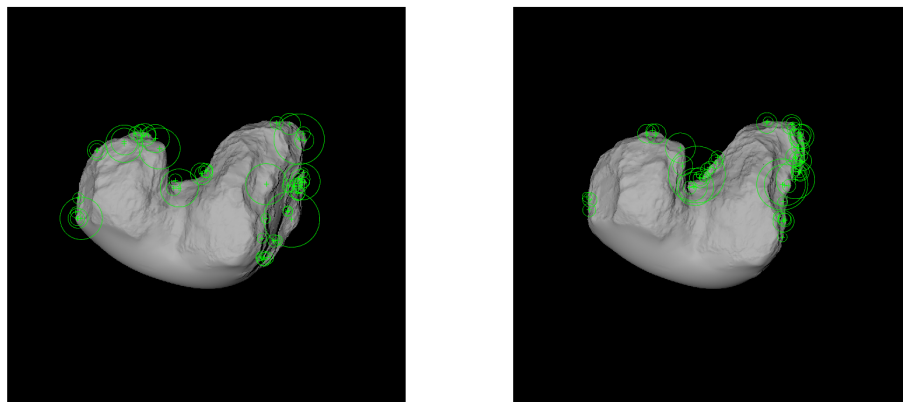


Fig. 9 SURF features extracted from POV-Ray generated images

This proves that an image collection based on the DRL policy results in a better capability of reconstructing the body shape and being compliant with the storage memory requirements. Nevertheless, the body shape is still far from being completely reconstructed, because of the scarcity of salient features in the generated images and because of simplified IP adopted. Future work shall investigate the possibility of reconstructing the body shape exploiting a previous model registration. Moreover, the effect of navigation uncertainties on the mapping process shall be accounted for, verifying the reconstruction capabilities with a more complex IP algorithm, that does not assume an exact knowledge of the relative pose.

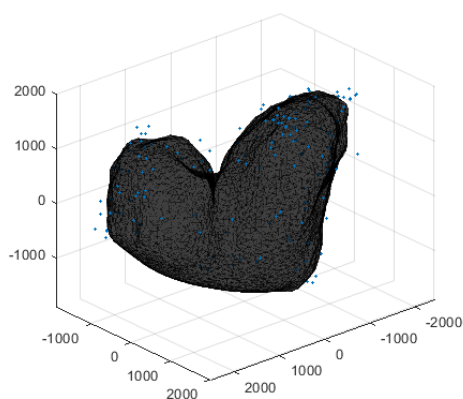


Fig. 10 UNI policy point cloud

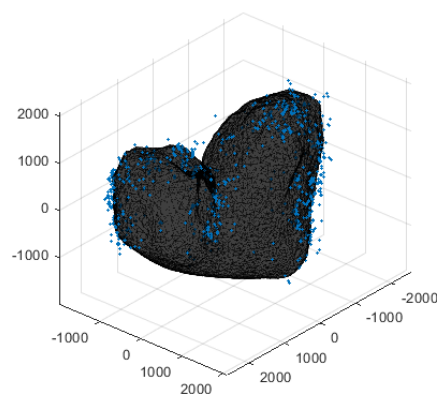


Fig. 11 DQN policy point cloud

VII. Conclusion

Small bodies mapping is a crucial but challenging capability for space exploration missions. Current missions heavily rely on human intervention not only for on ground map refinement, but also for operations supervision and planning. In fact, the extreme variety of body shapes, dynamical environments and illumination conditions makes spacecraft autonomous mapping a complex task in relation to the limited computational resources available on-board. This paper develops a method to autonomously plan the timing of observations during the mapping of an unknown small body, with particular application to imaging for SPC. The goal is to define a policy that improves mapping quality, while both limiting the amount of images to downlink and fastening the mapping process. The planning framework is defined as a POMDP, proposing a novel problem architecture focused on data collection. DRL is exploited to design the planning policies. The proposed approach enhances the imaging efficiency with respect to the benchmark solution.

The policy performance is verified by means of point cloud generation with a feature detection and matching IP, thus verifying the effectiveness of the hand-designed rewards used during the learning process. Two results are consequent to the application of the DRL-based policy: an increased performance in the point cloud reconstruction and a correct handling of memory storage during the mapping.

References

- [1] Gaskell, R. W., Barnouin-Jha, O. S., Scheeres, D. J., Konopliv, A. S., Mukai, T., Abe, S., Saito, J., Ishiguro, M., Kubota, T., Hashimoto, T., Kawaguchi, J., Yoshikawa, M., Shirakawa, K., Kominato, T., Hirata, N., and Demura, H., "Characterizing and navigating small bodies with imaging data," *Meteoritics and Planetary Science*, Vol. 43, No. 6, 2008, pp. 1049–1061. <https://doi.org/10.1111/j.1945-5100.2008.tb00692.x>.
- [2] Giese, B., Oberst, J., Kirk, R., and Zeitler, W., "The topography of asteroid Ida: A comparison between photogrammetric and two-dimensional photoclinometric image analysis," *International Archives of Photogrammetry and Remote Sensing*, Vol. 31, 1996, p. B3.
- [3] de Santayana, R. P., and Lauer, M., "Optical measurements for rosetta navigation near the comet," *Proceedings of the 25th International Symposium on Space Flight Dynamics (ISSFD)*, Munich, 2015.
- [4] Feder, H. J. S., Leonard, J. J., and Smith, C. M., "Adaptive mobile robot navigation and mapping," *The International Journal of Robotics Research*, Vol. 18, No. 7, 1999, pp. 650–668.
- [5] Carrillo, H., Reid, I., and Castellanos, J. A., "On the comparison of uncertainty criteria for active SLAM," *2012 IEEE International Conference on Robotics and Automation*, IEEE, 2012, pp. 2080–2087.
- [6] Kollar, T., and Roy, N., "Trajectory optimization using reinforcement learning for map exploration," *The International Journal of Robotics Research*, Vol. 27, No. 2, 2008, pp. 175–196. <https://doi.org/10.1177/0278364907087426>.
- [7] Agha-mohammadi, A.-a., Chakravorty, S., and Amato, N. M., "FIRM : Sampling-based Feedback Motion Planning Under Motion Uncertainty and Imperfect Measurements," *The International Journal of Robotics Research*, Vol. 33, No. 2, 2003, pp. 268–304. <https://doi.org/10.1177/0278364913501564>.
- [8] Olson, C. G., "Sequential Estimation Methods for Small Body Optical Navigation," Ph.D. thesis, The University of Texas at Austin, 2016.
- [9] Pesce, V., Agha-mohammadi, A.-a., and Lavagna, M., "Autonomous navigation & mapping of small bodies," *2018 IEEE Aerospace Conference*, IEEE, 2018, pp. 1–10. <https://doi.org/10.1109/AERO.2018.8396797>.
- [10] Chan, D. M., and Agha-mohammadi, A.-a., "Autonomous Imaging and Mapping of Small Bodies Using Deep Reinforcement Learning," *2019 IEEE Aerospace Conference*, 2019.
- [11] Piccinin, M., Pesce, V., Silvestrini, S., and Lavagna, M., "Smart Autonomous Imaging Plan For Small Bodies Efficient Mapping," *15th Symposium on Advanced Space Technologies in Robotics and Automation*, 2019.
- [12] Hartley, R., and Zisserman, A., *Multiple view geometry in computer vision*, Cambridge university press, 2003.
- [13] de Santayana, R. P., Lauer, M., Muñoz, P., and Castellini, F., "Surface Characterization and Optical Navigation at the Rosetta Flyby of Asteroid Lutetia," *International Symposium on Space Flight Dynamics*, 2014.
- [14] Sutton, R. S., and Barto, A. G., *Reinforcement learning: An introduction*, MIT press, 2018.
- [15] Shani, G., Pineau, J., and Kaplow, R., "A survey of point-based POMDP solvers," *Autonomous Agents and Multi-Agent Systems*, Vol. 27, No. 1, 2013, pp. 1–51. <https://doi.org/10.1007/s10458-012-9200-2>.
- [16] Kaelbling, L. P., Littman, M. L., and Cassandra, A. R., "Planning and acting in partially observable stochastic domains," *Artificial intelligence*, Vol. 101, No. 1-2, 1998, pp. 99–134. [https://doi.org/10.1016/S0004-3702\(98\)00023-X](https://doi.org/10.1016/S0004-3702(98)00023-X).
- [17] Lin, L., and Mitchell, T., "Memory Approaches to Reinforcement Learning in Non-Markovian Domains," Tech. rep., Pittsburgh, PA, USA, 1992.
- [18] Géron, A., *Hands-on machine learning with Scikit-Learn and TensorFlow: concepts, tools, and techniques to build intelligent systems*, O'Reilly Media, Inc., 2017.

- [19] Durrant-Whyte, H., and Bailey, T., "Simultaneous localization and mapping: part I," *IEEE robotics & automation magazine*, Vol. 13, No. 2, 2006, pp. 99–110. <https://doi.org/10.1109/MRA.2006.1638022>.
- [20] Mur-Artal, R., Montiel, J. M. M., and Tardos, J. D., "ORB-SLAM: a versatile and accurate monocular SLAM system," *IEEE transactions on robotics*, Vol. 31, No. 5, 2015, pp. 1147–1163. <https://doi.org/10.1109/TRO.2015.2463671>.
- [21] Mihaylova, L., Lefebvre, T., Bruyninckx, H., Gadeyne, K., and De Schutter, J., "A comparison of decision making criteria and optimization methods for active robotic sensing," *International Conference on Numerical Methods and Applications*, Springer, 2002, pp. 316–324. https://doi.org/10.1007/3-540-36487-0_35.
- [22] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al., "Human-level control through deep reinforcement learning," *Nature*, Vol. 518, No. 7540, 2015, p. 529. <https://doi.org/10.1038/nature14236>.
- [23] Saaty, T. L., "A scaling method for priorities in hierarchical structures," *Journal of Mathematical Psychology*, Vol. 15, No. 3, 1977, pp. 234–281. [https://doi.org/10.1016/0022-2496\(77\)90033-5](https://doi.org/10.1016/0022-2496(77)90033-5).
- [24] Riedmiller, M., "10 steps and some tricks to set up neural reinforcement controllers," *Neural networks: tricks of the trade*, Springer, 2012, pp. 735–757. https://doi.org/10.1007/978-3-642-35289-8_39.